Pre-eclampsia / Eclampsia

Focused History
Gestation, OB history, seizure duration/onset, pre-eclampsia symptoms

Pre-Eclampsia
SBP 140-170mmHg
DBP 90-110mmHg

- >20 weeks pregnancy
- Confusion
- Shortness of breath
- Vaginal bleeding

- Maintain quiet environment with minimal body movement
- Ensure position of comfort, semi-recumbent or left lateral position if in third trimester

If HR >60bpm
Administer Labetalol 20mg IV/2 minutes
Repeat every 10 minutes as required, max 80mg

- Administer to decrease MAP by 20-25% initially
- Target BP of 140/90

AP

Eclampsia

- SBP>160mmHg DBP>110mmHg
- >20 weeks gestation to up to 6 weeks post-partem
- Suspected pre-eclampsia with prodromal clinical features
- Seizure activity

Consider ALS Back-Up

- Position patient in **left lateral tilt**
- **Airway management** and suctioning if needed.
- High flow **O2 therapy** (10-15L/min via non-rebreather mask)
- Bilateral **large bore IV** access above diaphragm (16-18gauge)

1st line treatment
6g Magnesium Sulphate IV/IO
over 5-10 minutes
Use syringe driver or dilute in 100ml Normal Saline.
OR
8g Magnesium Sulphate IM
(4g in each gluteal area)

AP

Seizure Terminated?

NO

Diazepam OR Midazolam
IV: 5mg IV/IO slow push/2mins
IM: 10mg IM
(Repeat once if patient is still seizing after 5 minutes)

EP

Seizure Terminated?

EP

YES

Hypertension Management Post-Seizure
If SBP>160mmHg, DBP >110mmHg
Labetalol
20mg IV/IO over 2 minutes
(May repeat every 10 minutes (max 80mg total

Patient Care Goals

1. Recognize serious conditions associated with pregnancy and hypertension
2. Prevention of eclampsia-related seizures
3. Provide adequate treatment for eclampsia-related seizures

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Female patient, more than 20 weeks' gestation, presenting with hypertension and evidence of end-organ dysfunction, including renal insufficiency, liver involvement, neurological, or hematological involvement.
    - May occur up to 6 weeks postpartum, but is rare after 48 hours post-delivery
- Presenting symptoms of pre-eclampsia, including:
    - Severe features of pre-eclampsia include: (Pregnancy-induced hypertension)
    - Severe hypertension (SBP greater than 160, DBP greater than 110)
    - Headache
    - Confusion/altered mental status
    - Vision changes including blurred vision, spots/floaters, loss of vision (these symptoms are often a precursor to seizure)
    - Right upper quadrant or epigastric pain
    - Shortness of breath/Pulmonary edema
    - Ecchymosis suggestive of low platelets (bruising, petechiae)
    - Vaginal bleeding suggestive of placental abruption
    - Focal neurologic deficits suggesting hemorrhagic or thromboembolic stroke
- Presenting symptoms of eclampsia, including:
    - Any pregnant patient who is seizing should be assumed to have eclampsia and treated as such.
    - Seizure in any late term pregnancy or postpartum patient

Exclusion Criteria

- Non-pregnant patients with hypertension or seizures.

Key Patient Assessment Points

1. **Obtain Obstetric History**
    - Gestational age in weeks or recent post-partum
    - Symptoms suggestive of end organ involvement such as headache, confusion, visual disturbances, seizure, epigastric pain, right upper quadrant pain, nausea/vomiting, stroke symptoms, shortness of breath
    - Previous history of hypertension or known pre-eclampsia
2. **Focused Obstetric Assessment**
    - Constitutional: vital signs, skin color
    - Abdomen: distension, tenderness, uterine rigidity
    - Genitourinary: visible bleeding
    - Neurologic: mental status, focal deficits
3. **Determine Triage Category**
    - Determine Triage Category as per Patient Triage Categories CPG.
        - Red Priority:
            - Eclampsia / Seizures in pregnancy
            - Severe Hypertension (SBP greater than 160 or DBP greater than 110).

Patient Management & Interventions

1. **Primary and Secondary Survey**
    - Perform primary and secondary survey as per Universal Care CPG.
2. **Consider ALS back-up**
    - Consider the requirement for ALS back-up as per Universal Care CPG.
    - ALS back-up should be routinely requested for:

    - Eclampsia
    - Pre-eclampsia with severe hypertension
3. **Administer Oxygen**
    - 10-15Lpm via NRB mask
    - Titrate oxygen to maintain Sp02 > 94%
4. **Establish IV Access**
    - Consider IO access if IV access cannot be easily gained.
5. **Position Patient**
    - Position the patient in the left lateral tilt position.
6. **Pre-Eclampsia without Severe Hypertension**
    - Inclusion Criteria:
        - SBP 140-170mmHg, DBP 90-110mmHg
    - Management:
        - Maintain quiet environment with minimal body movement
        - Ensure position of comfort, semi-recumbent or left lateral position if in second or third trimester
7. **Pre-Eclampsia with Severe Hypertension**
    - Management Inclusion Criteria:
        - Severe hypertension (SBP greater than 160 or DBP greater than 110) lasting more than 15 min with associated pre-eclampsia symptoms
    - **Management Goals:**
        - Severely elevated blood pressures must be treated to reduce the risk of maternal stroke.
        - However, goal blood pressure should be roughly 140/90 to maintain uterine perfusion and to keep the fetus well-oxygenated
        - The goal of BP is approximately 140/90 to reduce stroke risk but maintain uterine perfusion.
    - **Administer:**
        - Labetalol or Magnesium Sulphate.
        - The routine administration of Magnesium Sulphate in Pre-eclampsia is not recommended.
        - **Labetalol:** 20 mg IV, over 2 minutes
            - May repeat every 10 minutes for persistent severe hypertension with pre-eclampsia symptoms (maximum 80mg total)
            - The goal is to reduce MAP by 20-25% initially
            - Ensure that HR is greater than 60 BPM before administration
        - Magnesium Sulphate: 4 g IV over 5-10 min, followed by 2 g/hr
            - Reassess vital signs every 10 minutes during transport.
8. **Eclampsia**
    - Management Inclusion Criteria:
        - Seizures associated with pregnancy greater than 20 weeks of gestation
    - **First-line intervention:**
        - **Administer Magnesium Sulphate:**
            - Magnesium Sulphate: 6 g IV over 5-10 minutes OR
            - Magnesium Sulphate: 8 g IM (4 grams in each gluteal area).
        - IV administration is preferred if IV access is available.
        - IV Magnesium Sulphate should be administered via syringe driver when available. However, dilution in 100 mL of Normal Saline is acceptable.
    - **Refractory seizures or Magnesium Sulphate is not available:**
        - **Administer Benzodiazepines**
        - Diazepam OR Midazolam
            - IV: 5mg IV/IO push over 2 minutes
            - IM: 10mg IM
        - May be repeated once if the patient is still seizing after 5 minutes.
9. **Patient Reassessment**

- Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Patients in the second or third trimester of pregnancy should be transported on the left side or with the uterus manually displaced to the left to ensure adequate uterine perfusion

Clinical Insights & Additional Guidance

- Delivery of the placenta is the only definitive management for pre-eclampsia and eclampsia.
- Early treatment of severe pre-eclampsia with Magnesium for seizure prophylaxis and anti-hypertensives significantly reduces the rate of eclampsia. The use of Magnesium is encouraged if signs of severe pre-eclampsia are present to prevent seizures.
- Patients with a history of chronic hypertension may have superimposed pre-eclampsia.
- Although less frequent, eclampsia, including eclampsia-related seizures, can occur in postpartum patients.
- Magnesium toxicity
    - Magnesium toxicity may develop with magnesium administration. The progression of magnesium toxicity symptoms is:
        - Hypotension followed by
        - Loss of deep tendon reflexes followed by
        - Somnolence, slurred speech followed by
        - Respiratory paralysis followed by
        - Cardiac arrest
    - Treatment of Magnesium toxicity
        - Stop the Magnesium drip
        - Give calcium gluconate 3 g IV or calcium chloride 1 g IV in cases of pending respiratory arrest (if available)
        - Support respiratory effort

Documentation

- Identification: Document patient's age, gestational age (in weeks or recent postpartum status), and any known history of hypertension or pre-eclampsia.
- Mechanism: Record the onset and duration of symptoms (e.g., hypertension, seizures, headache) and any associated events (e.g., trauma, placental abruption).
- Injuries/Illness: Note specific findings such as severe hypertension (SBP >160, DBP >110), neurological changes (confusion, vision loss), respiratory distress, or signs of end-organ involvement (e.g., epigastric pain, ecchymosis).
- Signs/Symptoms: Document vital signs (including repeat BP every 5-10 minutes), level of consciousness, deep tendon reflexes, and response to treatment (e.g., magnesium sulfate, labetalol).
- Treatment: Record all interventions, including magnesium sulfate administration (dose, route, time), antihypertensive therapy (e.g., labetalol 20 mg IV), IV fluid rates (max 80 mL/hr), oxygen administration, and any benzodiazepine use for seizures.
- Allergies: Note any known allergies to medications (e.g., magnesium, benzodiazepines).
- Medications: List current or recent medications, including prenatal vitamins or antihypertensives.
- Background: Include obstetric history (e.g., previous pregnancies, complications) and current pregnancy details.
- Other Information: Document transport details (left lateral position, ALS back-up requested), patient safety considerations (e.g., magnesium toxicity signs), and communication with receiving facility.

Key Performance Indicators

1. Administer antihypertensive treatment (e.g. Labetalol) for patients with severe hypertension(SBP >160 or DBP >110).
2. Position second and third trimester patients with left lateral tile or manual uterine displacement.

References

1. Ambulance Victoria (2022) *Clinical Practice Guidelines for Ambulance and MICA Paramedics*. Melbourne: Ambulance Victoria.
2. American College of Obstetricians and Gynecologists (2020) 'Practice Bulletin No. 222: Gestational Hypertension and Preeclampsia', *Obstetrics & Gynecology*, 135(6), pp. e237–e260.
3. Magee, L.A. et al. (2021) 'The 2021 International Society for the Study of Hypertension in Pregnancy Guidelines on the Prevention and Treatment of Preeclampsia', *Hypertension*, 77(5), pp. 1423–1436.
4. National Association of State EMS Officials (2023) *National Model EMS Clinical Guidelines*. Version 3. Washington, DC: NASEMSO.
5. National Institute for Health and Care Excellence (2019) *Hypertension in Pregnancy: Diagnosis and Management* [NG133]. London: NICE.
6. Pre-Hospital Emergency Care Council (2017) *Clinical Practice Guidelines for Pre-Hospital Emergency Care*. Naas: PHECC.
7. Queensland Ambulance Service (2023) *Clinical Practice Guidelines: Pre-eclampsia/Eclampsia*. Brisbane: Queensland Government.
8. South Australian Ambulance Service (2025) *Clinical Practice Guidelines: Pre-eclampsia/Eclampsia. Adelaide: SA Ambulance Service.*
9. *World Health Organization (2011) WHO Recommendations: Prevention and Treatment of Pre-eclampsia and Eclampsia. Geneva: World Health Organization.*

Mass Casualty Incident Triage